

# A Practical Guide to Stacking ensemble

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TOPIC -- STACKING ENSEMBLE

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Medicine Ensemble classifiers have been successfully applied in neuroscience, proteomics and medical diagnosis like in neuro-cognitive disorder (i.e. Alzheimer or myotonic dystrophy) detection based on MRI datasets, cervical cytology classification. Besides, ensembles have been successfully applied in medical segmentation tasks, for example brain tumor and hyperintensities segmentation.

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Stacking (sometimes called stacked generalization) involves training a model to combine the predictions of several other learning algorithms. First, all of the other algorithms are trained using the available data, then a combiner algorithm (final estimator) is trained to make a final prediction using all the predictions of the other algorithms (base estimators) as additional inputs or using cross-validated predictions from the base estimators which can prevent overfitting. If an arbitrary combiner algorithm is used, then stacking can theoretically represent any of the ensemble techniques described in this article, although, in practice, a logistic regression model is often used as the combiner. Stacking typically yields performance better than any single one of the trained models. It has been successfully used on both supervised learning tasks (regression, classification and distance learning ) and unsupervised learning (density estimation). It has also been used to estimate bagging's error rate. It has been reported to out-perform Bayesian model-averaging. The two top-performers in the Netflix competition utilized blending, which may be considered a form of stacking. Bayesian predictive stacking generalizes the idea of stacking from the statistical estimation literature to the combination of posterior predictive distributions. These ideas have also been developed and investigated for Gaussian process models, especially for spatial data analysis and can be used to construct transfer learning frameworks for massive spatial data sets.

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The Bayes optimal classifier is a classification technique. It is an ensemble of all the hypotheses in the hypothesis space. On average, no other ensemble can outperform it. The Naive Bayes classifier is a version of this that assumes that the data is conditionally independent on the class and makes the computation

more feasible. Each hypothesis is given a vote proportional to the likelihood that the training dataset would be sampled from a system if that hypothesis were true. To facilitate training data of finite size, the vote of each hypothesis is also multiplied by the prior probability of that hypothesis. The Bayes optimal classifier can be expressed with the following equation:

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Financial decision-making Ensemble learning methods have been widely adopted in finance for tasks such as credit scoring, bankruptcy prediction, and risk management. By combining multiple base models, ensembles can exploit nonlinear relationships, handle high-dimensional and noisy data, and often deliver more stable out-of-sample performance than single models or traditional statistical baselines such as logistic regression and linear factor models. This approach aligns with the broader trend in financial machine learning described by Marcos López de Prado, who argues that combining diverse learners can improve robustness, mitigate overfitting, and extract persistent patterns from noisy financial time series. In retail and corporate credit scoring, ensemble classifiers such as random forests, gradient boosting machines and stacked models are frequently used to assess default risk. A recent systematic literature review of machine-learning credit-scoring studies published between 2018 and 2024 reports that tree-based ensembles and boosting methods are among the most commonly applied techniques and typically achieve higher predictive accuracy than traditional scorecards or single classifiers. Ensemble methods have also been applied to corporate failure and bankruptcy prediction. Studies comparing different ensemble constructions (such as bagging, boosting and heterogeneous classifier pools) report that well-tuned ensembles tend to achieve higher classification accuracy and more robust performance across industries than individual models. The accuracy of prediction of business failure is a very crucial issue in financial decision-making. Therefore, different ensemble classifiers are proposed to predict financial crises and financial distress. Also, in the trade-based manipulation problem, where traders attempt to manipulate stock prices by buying and selling activities, ensemble classifiers are required to analyze the changes in the stock market data and detect suspicious symptom of stock price manipulation. Implementation of ensemble learning not only with machine learning models but also with mathematical frameworks such as hidden Markov models for applications in trading, regime detection, and asset pricing remains a rapidly developing area of research.

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Cross-Validation Selection can be summed up as: "try them all with the training set, and pick the one that works best". Gating is a generalization of Cross-Validation Selection. It involves training another learning model to decide which of the models in the bucket is best-suited to solve the problem. Often, a perceptron is used for the gating model. It can be used to pick the "best" model, or it can be used to give a linear weight to the predictions from each model in the bucket. When a bucket of models is used with a large set of problems, it may be desirable to avoid training some of the models that take a long time to train. Landmark learning is a meta-learning approach that seeks to solve this problem. It involves training only the fast (but imprecise) algorithms in the bucket, and then using the performance of these algorithms to help determine which slow (but accurate) algorithm is most likely to do best.

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For each model  $m$  in the bucket: Do  $c$  times: (where ' $c$ ' is some constant)  
Randomly divide the training dataset into two sets:  $A$  and  $B$  Train  $m$  with  $A$  Test  $m$  with  $B$  Select the model that obtains the highest average score

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Implementations in statistics packages R: Several packages offer Bayesian model averaging tools, including the BMS (an acronym for Bayesian Model Selection) package, the BAS (an acronym for Bayesian Adaptive Sampling) package, and the BMA package. Other packages implementing predictive stacking for Gaussian process models include the spStack package. Python: scikit-learn, a package for machine learning in Python offers packages for ensemble learning including packages for bagging, voting and averaging methods. MATLAB: classification ensembles are implemented in Statistics and Machine Learning Toolbox.

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Bayesian model averaging Bayesian model averaging (BMA) makes predictions by averaging the predictions of models weighted by their posterior probabilities given the data. BMA is known to generally give better answers than a single model, obtained, e.g., via stepwise regression, especially where very different models have nearly identical performance in the training set but may otherwise perform quite differently. The question with any use of Bayes' theorem is the prior, i.e., the probability (perhaps subjective) that each model is the best to use for a given purpose. Conceptually, BMA can be used with any prior. R packages ensembleBMA and BMA use the prior implied by the Bayesian information criterion, (BIC), following Raftery (1995). R package BAS supports the use of the

priors implied by Akaike information criterion (AIC) and other criteria over the alternative models as well as priors over the coefficients. The difference between BIC and AIC is the strength of preference for parsimony. BIC's penalty for model complexity is  $\ln(n)k$ , while AIC's is  $2k$ . Large-sample asymptotic theory establishes that if there is a best model, then with increasing sample sizes, BIC is strongly consistent, i.e., will almost certainly find it, while AIC may not, because AIC may continue to place excessive posterior probability on models that are more complicated than they need to be. On the other hand, AIC and AICc are asymptotically "efficient" (i.e., minimum mean square prediction error), while BIC is not. Haussler et al. (1994) showed that when BMA is used for classification, its expected error is at most twice the expected error of the Bayes optimal classifier. Burnham and Anderson (1998, 2002) contributed greatly to introducing a wider audience to the basic ideas of Bayesian model averaging and popularizing the methodology. The availability of software, including other free open-source packages for R beyond those mentioned above, helped make the methods accessible to a wider audience.